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Intra-arc tectonic deformation around the Irazú–Turrialba volcanic complex, Costa Rica

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ABSTRACT

The tectonic setting of the Irazú–Turrialba volcanic complex in central Costa Rica is characterised by a diverse tectonic interplay, including oblique subduction of the Cocos Plate beneath the Caribbean Plate and a diffuse region of deformation known as the Central Costa Rica Deformed Belt. The western (inferred) limits of this deformation region consist of the northwest propagation of a tectonic forearc sliver along the strike of the Central America Volcanic Arc. The eastern limit is attributed to the northeast escape of the overriding Panama Microplate. We exploit the recordings of a dedicated seismic network composed of 42 stations, operating from April 2018 to March 2019, to investigate this complex deformation associated with volcanic systems. Our local network improved the magnitude of completeness and allowed us to invert focal mechanisms that revealed the kinematics of tectonic and magmatic structures. We detected and located 967 local earthquakes, and constrained a new, local 1D velocity model for the investigated volcanic region. The relocated seismicity and associated focal mechanisms constrained for major earthquakes point out local crustal deformation, obliquely-orientated from the major regional tectonic blocks. We propose a conceptual geological model suggesting that deformation occurring at each volcano is subject to distinct tectonic regimes. In particular, we suggest that Irazú is sensitive to motion accumulated from the forearc sliver, while Turrialba is affected by the regional subduction.

1. Introduction

Intra-arc deformation plays a key role in accommodating upper-plate strain along volcanic arcs and in controlling magma ascent, storage and eruption style (Tibaldi, 1992; Tibaldi et al., 2009). The Irazú–Turrialba Volcanic Complex (ITVC) in central Costa Rica sits within the Central Costa Rica Deformed Belt (CCRDB) (Fig. 1), a broad zone of crustal deformation where regional plate motions and arc magmatism strongly interact (Jacob et al., 1991; Molnar and Sykes, 1969; Stoiber and Carr, 1973; Marshall et al., 2000; Sallarès et al., 2000; Norabuena et al., 2004; Fisher et al., 2004; Lewis et al., 2008; Van Benthem and Govers, 2010; Kobayashi et al., 2014). The ITVC marks the southeastern terminus of the Central Costa Rica Volcanic Front and coincides with the transition from an active volcanic arc to the amagmatic, uplifted Talamanca Cordillera. Despite intensive work on the regional tectonics (e.g., Alvarado et al., 2006) and on the magmatic plumbing system of the ITVC (Jiwani-Brown et al., 2022, 2024), our understanding of the local-scale kinematics within the CCRDB beneath and around the ITVC need to be improved. Previous geodetic, geologic and seismological studies have shown that the CCRDB accommodates a significant fraction of the upper-plate deformation induced by subduction of the Cocos Plate and collision of the Cocos Ridge, expressed as right- and left-lateral strike-slip faulting, thrusting and transpression at different scales (Marshall et al., 2000; Lewis et al., 2008; Van Benthem and Govers, 2010; Kobayashi et al., 2014; Carvajal-Soto et al., 2020; López-Saborío and Carvajal-Soto, 2024). Within this setting, the Atirro–Río Sucio fault system (Fig. 2a) and associated structures define a complex intra-arc fault network on which both tectonic and volcano-tectonic earthquakes occur.

Understanding the complex kinematics of the ITVC requires bridging the gap between large-scale plate interactions and the localised expression of crustal strain. The ITVC is an active twin-volcanic system, situated within 50 km of Costa Rica's capital city, San Jose. The summits of each volcano are 10 km apart and form a complex that is 60 km by 45 km at the base. This study implements

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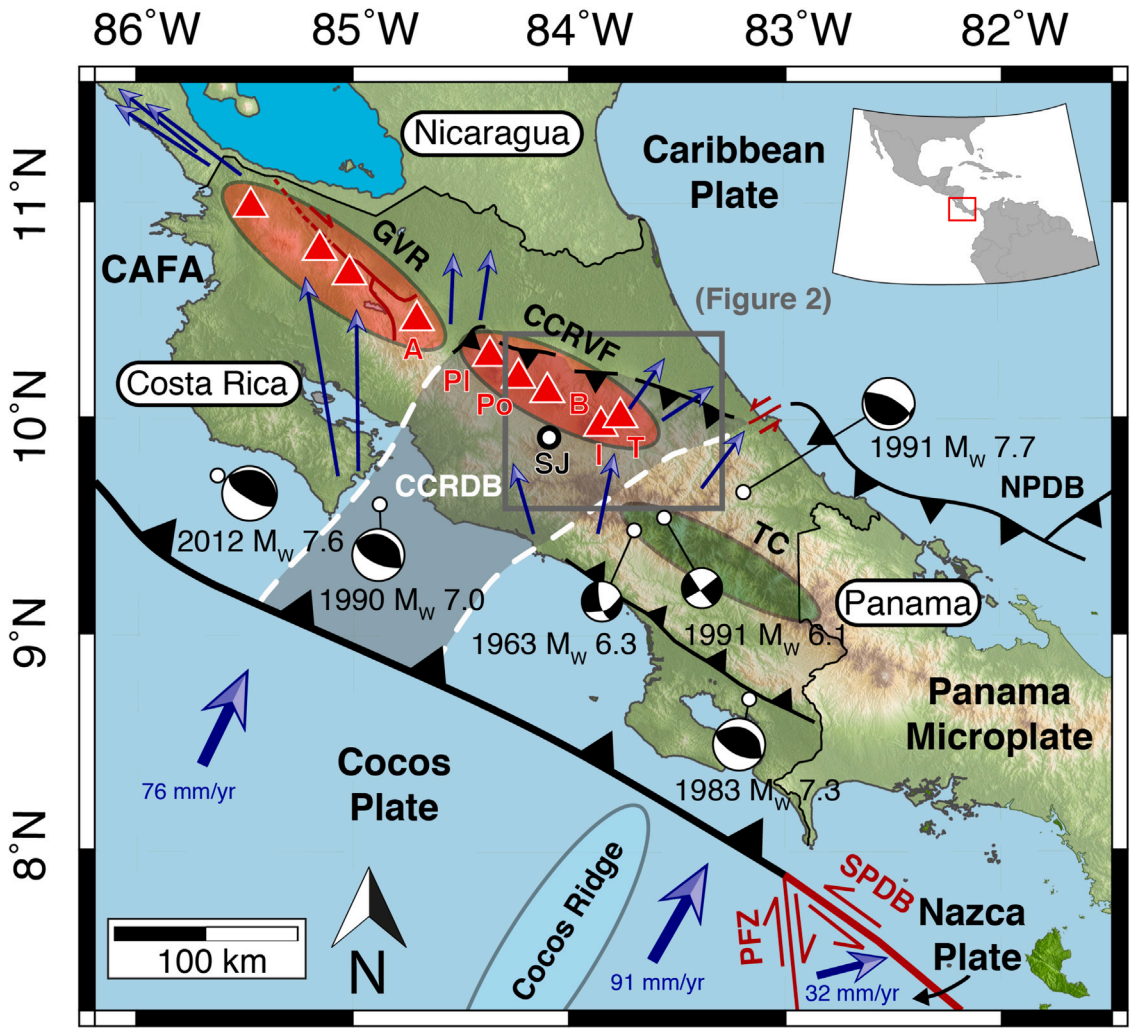


Fig. 1. Tectonic Map of Costa Rica and surrounding regions. Regional-scale major tectonic features for central Costa Rica. The key faults shown are: North Panama Deformation Boundary (NPDB), South Panama Deformation Boundary (SPDB) and Panama Fracture Zone (PFZ). The white dashed line represents the potential region of the Central Costa Rica Deformation Belt (CCRDB) and the grey square represents the region presented in Fig. 2. The Holocene volcanic edifices (red triangles) include the Guanacaste Volcanic Range (GVR) and Central Costa Rica volcanic front (CCRVF). The marked volcanoes are Arenal (A), Platanar (Pl), Poás (Po), Barva (B), Irazú (I) and Turrialba (T). The main alpine segment is marked as the Talamanca Cordillera (TC). The capital city of San Jose (SJ) is also indicated. Key tectonic features adapted from Alvarado et al. (2017), Montero et al. (2013b) and Denyer (2009).

a dense, temporary network of 42 seismic stations deployed across the ITVC to produce a high-resolution portrait of its seismo-tectonic framework. A prerequisite for this analysis is the development of a new, robust minimum 1D P-wave velocity model for the volcanic massif, a recognised necessity for improving earthquake locations in complex volcanic regions (Kissling et al., 1994). Using this model, we apply the double-difference relocation algorithm (HypoDD) to a newly compiled earthquake catalogue, followed by fault-plane analysis for key seismic events (Waldhauser and Ellsworth, 2000). Through this multi-faceted approach, we constrain local-scale faulting and stress orientations, ultimately allowing us to establish a detailed mechanical interpretation of how the ITVC responds to and participates in the multi-scale tectonic dynamics of the CCRDB.

The interplay between tectonics and magmatism is fundamental to the evolution of volcanic arcs, with each system profoundly influencing the other. On a regional scale, the architecture of the overriding plate and the segmentation of the subducting slab dictate the location and geochemical signature of volcanic centres like the ITVC (Carr, 1984; Gazel et al., 2021). Locally, crustal fault systems act as primary conduits for magma ascent. For instance, transtensional tectonics associated with the Atirro-Río Sucio fault system have created a pull-apart basin that accommodates the ITVC, a structural setting highly favourable for magmatism (Montero et al., 2013a). This localised extension is further expressed by graben structures within the volcanic edifices themselves (Soto, 1988; Calvo et al., 2019). The dynamic influence of regional tectonics is also critical; large tectonic events, such as the 2012 M_W 7.6 Nicoya earthquake, have been shown to trigger seismic swarms at the ITVC by transferring stress to the magmatic-hydrothermal system (Lupi et al., 2014). Conversely, the presence and movement of magma actively shape local tectonics. The emplacement of magma reservoirs and conduits creates zones of crustal weakness and generates localised stress fields that can modify the regional pattern (Sparks et al., 2019). The migration and pressurisation of magmatic and hydrothermal fluids is a principal driver of local volcano-tectonic (VT) seismicity, where fluid pressure promotes rock failure on pre-existing faults (Neuberg, 2011; White and McCausland, 2016; Muller et al., 2024). At Turrialba, recent eruptive activity has been directly linked to seismic swarms driven by fluid migration through permeable domains (Conde et al., 2014; van der Laat et al., 2022), a process whose mechanics have been further illuminated by the analysis of Long-Period (LP) events (Eyre et al., 2015; Zecevic et al., 2016). Recent ambient-noise and local-earthquake tomography revealed low-velocity anomalies interpreted as mid-crustal reservoirs and mush zones beneath the complex

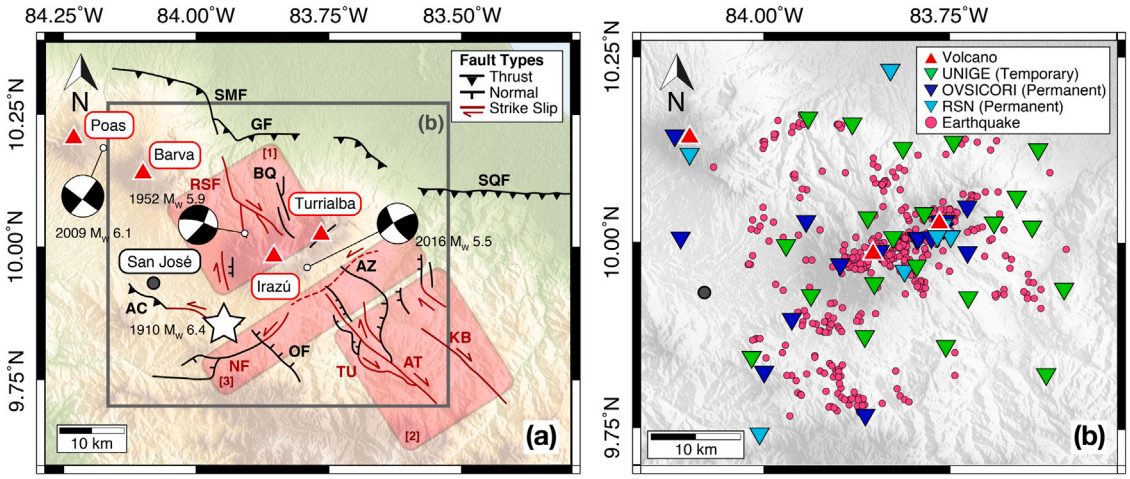


Fig. 2. Tectonic Map of th Irazú–Turrialba volcanic complex. (a) Local-scale major tectonic features of the Irazú–Turrialba Volcanic Complex. Montero et al. (2013b) separates the ITVC fault system into three principal segments; (1) the Río-Sucio segment, including the Río Sucio (RSF) and Blanquito (BQ) faults to the north-west, (2) Atirro–Tucurrique segment, including the Atirro (AT), Tucurrique (TU), Azul (AZ) and Kabebeta (KB) fault to the south-east, and (3) a central pull-apart basin accommodated within the ITVC and along the Navarro (NF), Orosi (OF) and Aguacaliente (AC) faults. **(b)** Network distribution and coverage of the temporary UNIGE stations (green triangles), and permanent OVSICORI (dark-blue triangles) and RSN (sky-blue triangles) stations, data recorded between April 2018 to March 2019. The earthquakes shown in dark-pink are the ones used for the velocity model inversions with *Velest* and *HypoInverse*. Key tectonic features adapted from Alvarado et al. (2017), Montero et al. (2013b) and Denyer (2009).

(Jiwani-Brown et al., 2022, 2024), but the detailed geometry of brittle structures accommodating intra-arc strain remains debated.

High-resolution earthquake relocations and focal mechanisms provide an opportunity to bridge this gap between regional deformation and local magmatic plumbing. Double-difference techniques such as *HypoDD* can collapse diffuse seismic clouds onto clusters that can be more discretely interpreted as tectonic structures. This, combined with robust velocity models, resolve fault geometries and stress orientations within the brittle crust (Waldhauser and Ellsworth, 2000). While these methods have been applied along the Costa Rican subduction interface, they have yet to be systematically used to characterise shallow intra-arc deformation in the CCRDB.

The goal of this study is to quantify the local seismo-tectonic deformation around the ITVC and to assess how it relates to both the CCRDB stress field and the magmatic system. We use an 11-month deployment of a dense temporary seismic network, complemented by permanent stations, to detect and relocate local earthquakes, derive a new minimum 1D velocity model, and compute focal mechanisms. We then integrate these results with existing tomographic images and regional tectonic models to propose a conceptual framework for intra-arc deformation at the ITVC.

2. Geological setting

2.1. Regional tectonic framework and the CCRDB

The regional tectonics of Costa Rica are governed by oblique subduction of the Cocos Plate beneath the Caribbean Plate at rates of 75 to 80 mm yr^{-1} (Argus et al., 2011; Carvajal-Soto et al., 2020). Subduction of the buoyant Cocos Ridge strongly modifies slab geometry and upper-plate deformation, driving the northwestward escape of the Central America Forearc Sliver (CAFA) and the east–northeast motion of the Panama Microplate (LaFemina et al., 2009; Staller et al., 2016; Van Benthem and Govers, 2010; Kobayashi et al., 2014; Montero et al., 2017). It has also been argued that oblique convergence and high mechanical coupling at the longitude of the Nicoya Peninsula drive strain partitioning, can promote the development of the CAFA and its northwestward motion. Both mechanisms likely contribute (LaFemina et al., 2009), requiring trailing edge deformation that is typically expressed as diffuse boundary zones such as the CCRDB. The collision of the Cocos Ridge is the primary driver of upper-plate shortening and lateral extrusion in the western Caribbean region, with deformation accommodated along the Middle America Trench, the Fila Costeña thrust belt and the North Panama Deformed Belt (Van Benthem and Govers, 2010; Carvajal-Soto et al., 2020; Bourke et al., 2023).

The boundary between the CAFA and the Panama Microplate, the CCRDB forms a 100-km-wide, diffuse zone of active faulting and distributed strain in the overriding crust (Marshall et al., 2000; Montero et al., 2013b; Alvarado et al., 2017; Carvajal-Soto et al., 2020). The CCRDB marks the western boundary of the Panama Microplate and the eastern boundary of the CAFA. The internal deformation zone of the Caribbean Plate, where both right-lateral and left-lateral strike-slip faulting, transtension and localised thrusting are observed (Marshall et al., 2000; Lewis et al., 2008; Linkimer et al., 2018; López-Saborío and Carvajal-Soto, 2024) can also be accommodated by the CCRDB, While it can also be argued that this deformation could reach as far as the Fila Costeña to Talamanaca region and into western Panama, focal mechanisms and GNSS data indicate that shallow (< 15 km) deformation within the CCRDB is dominated by strike-slip and transpressional regimes. Stress orientations are controlled by the combined effects of oblique subduction and ridge collision (LaFemina et al., 2009; Kobayashi et al., 2014; López-Saborío and Carvajal-Soto, 2024).

Large-scale seismicity can be attributed to a variation of processes, including plate boundary interactions and upper plate rupturing. Events located along the margin delineates a well-developed Benioff zone, with slab dips varying from 30° to 80° and becoming relatively shallow (~40°) beneath central Costa Rica and the ITVC (Protti et al., 1994; DeShon et al., 2006; Hayes et al., 2012; Arroyo et al., 2014). Notable historical events occurring in this region include the Patillos 1952 (M$_W$ 5.9) earthquake northwest of Irazú, the Cinchona 2009 (M$_W$ 6.1) earthquake on the flank of Poás volcano, and the Capellades 2016 (M$_W$ 5.5) earthquake located within the ITVC itself (Linkimer et al., 2018). Large subduction earthquakes (M$_W$ > 7), such as the Osa 1983, Limón 1991 and Nicoya 2012 events, occur at the plate interface and have been shown to trigger increased shallow seismicity within the CCRDB, including swarms beneath the ITVC (Montero-Pohly, 2021b; Tajima and Kikuchi, 1995; Linkimer Abarca et al., 2014; Lupi et al., 2014; Montero-Pohly, 2021a). Within central Costa Rica, additional strike-slip events (e.g. Buenavista 1983, Cuericí 1991) highlight the activity of upper-plate fault systems inside the CCRDB (Montero-Pohly, 2021b).

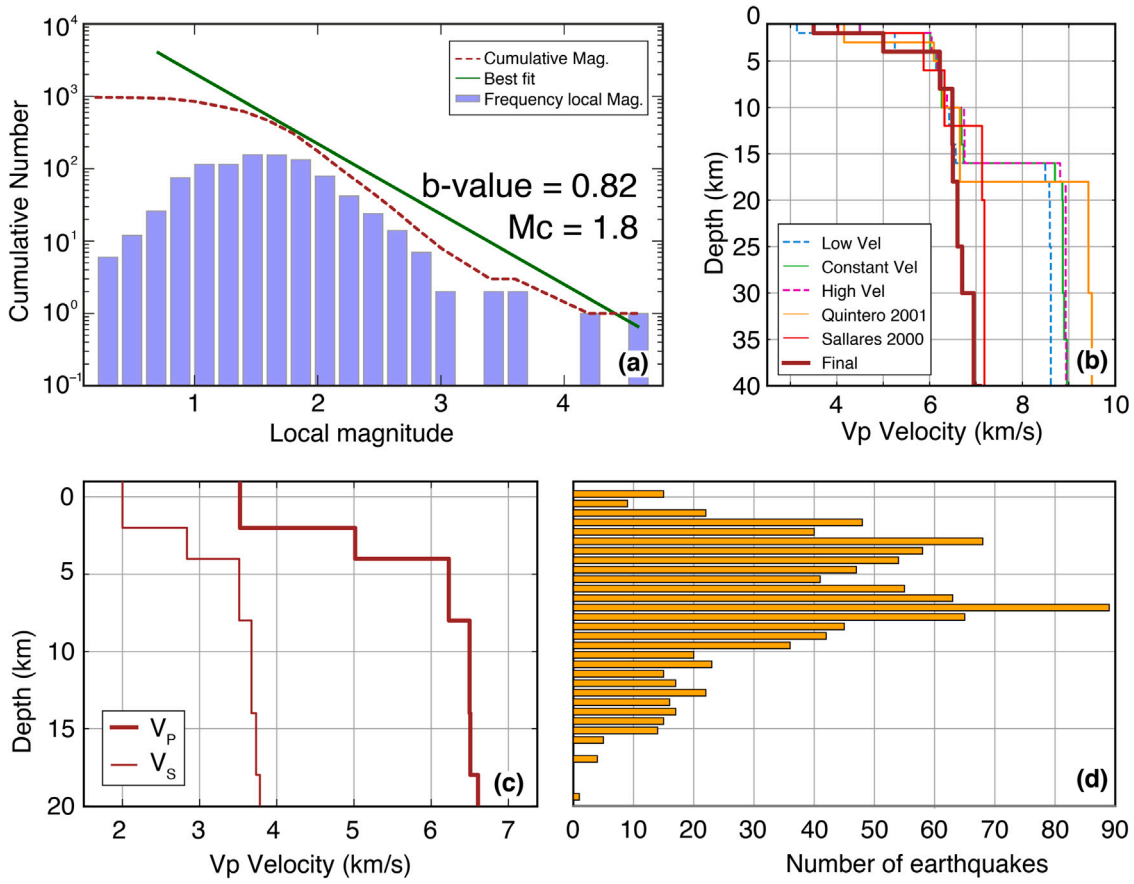


Fig. 3. Velocity model, b-value and earthquake distribution calculations. (a) B-values computed using the *GISMO* package (Thompson and Reyes, 2018). **(b)** Preliminary output velocity models from previous regional studies of Costa Rica (Sallarès et al., 2000; Quintero and Kissling, 2001), and high/low boundary models, **(c)** final computed output 1D velocity model for the ITVC, V_P values calculated with *HypoInverse*, and V_S using a V_P/V_S also computed using the *GISMO* package, **(d)** Distribution of earthquakes within depth. We note that after 17 km depth, the model becomes less constrained by a reduction in data.

Modern volcanism in Costa Rica defines a NW–SE oriented volcanic front, subdivided into the Guanacaste Volcanic Range, the Tilarán Volcanic Range (GVR) and the Central Costa Rica Volcanic Front (CCRVF), on which the ITVC lies (e.g. Gazel et al. (2021)). Subduction of the Cocos Ridge and associated Galapagos-related seamount tracks has produced some of the most trace-element enriched lavas worldwide and contributed to strong along-arc geochemical segmentation (Hoernle et al., 2008; Gazel et al., 2009, 2019). South of the ITVC, collision of the buoyant ridge and interaction with the Panama Microplate lead to termination of arc volcanism and uplift of the amagmatic Talamanca Cordillera (Gazel et al., 2019; Bourke et al., 2023).

The ITVC marks the southeastern end of the active Central American Volcanic Arc. Early work focused on the petrology and geochemistry of eruptive products (Alvarado et al., 2006; DeVitre et al., 2019), whereas recent geophysical studies have characterised its subsurface structure. Receiver-function analysis has imaged mid-crustal discontinuities and possible magmatic structures beneath Irazú (Villegas Alvarez et al., 2019). Ambient-noise tomography has revealed a low shear-velocity anomaly at ~6 km depth, interpreted as a mid-crustal reservoir or mush zone (Jiwani-Brown et al., 2022). Local-earthquake tomography shows contrasting V_P and V_S anomalies at 4–8 km depth beneath Irazú and Turrialba, suggesting that the shallow reservoirs of the two volcanoes are at least partly independent (Jiwani-Brown et al., 2024). Petrological constraints and magma mixing studies point to additional, deeper reservoirs at ~10–15 km depth (Alvarado et al., 2006; Di Piazza et al., 2015; de Moor et al., 2016; Di Piazza et al., 2019; DeVitre et al., 2019).

Seismic observations relating to volcanic processes indicate that both shallow and intermediate-depth seismicity are influenced by magmatic and hydrothermal processes. At Turrialba, swarms of volcano-tectonic (VT) and long-period (LP) events have been associated with fluid migration and changes in permeability within the upper crust, particularly during the 2009–2020 unrest phase (Conde et al., 2014; Eyre et al., 2015; Zecevic et al., 2016; van der Laat et al., 2022). Deformation studies and modelling of volume changes at 5–10 km depth further support the presence of an active magmatic system beneath the ITVC and its interaction with surrounding faults (Muller, 2018; Battaglia et al., 2019).

Within the CCRDB, the Atirro–Río Sucio fault system represents the main intra-arc shear zone in the vicinity of the ITVC (Montero et al., 2013a; Lupi et al., 2014). It is usually subdivided into three segments: (1) the Río Sucio segment to the northwest, (2) the Atirro–Tucurrique segment to the southwest, and (3) a central pull-apart basin that accommodates the ITVC along the Navarro, Orosi and Aguacaliente faults (Montero et al., 2013a,b; Calvo et al., 2019). Mapping and focal-mechanism analyses indicate a combination of right-lateral and left-lateral strike-slip faulting with localised normal faulting, consistent with a transtensional regime within the CCRDB (Montero et al., 2013b; Alvarado et al., 2017; Linkimer et al., 2018, 2021; López-Saborío and Carvajal-Soto, 2024).

Recent studies have proposed additional structures, such as the Liebres fault, inferred from seismicity patterns at 5–15 km depth and from deformation modelling (Soto et al., 2009; Linkimer et al., 2018; Muller, 2018; Battaglia et al., 2019). These structures likely provide pathways for magmatic and hydrothermal fluids and may link the ITVC to the broader Atirro–Río Sucio system. Rheological contrasts between

magmatic reservoirs and surrounding crust modulate the local stress field and intra-arc deformation (Cailleau et al., 2007; Cashman et al., 2017; Acocella, 2021). The existence of the ITVC magmatic system could be responsible for a mechanically weak segment of the upper plate, also demonstrated by Cailleau et al. (2007) to be present within the CAVA during the CAFA motion.

3. Methods

3.1. Seismic network

We deployed 20 broadband seismic stations in a network surrounding the ITVC (Fig. 2b). The sensors were three-component Nanometrics Trillium Compacts (20 s and 120 s). Our temporary network ran for a period of eleven months from deployment in April 2018 to removal in March 2019. All stations were equipped with Omnirecs DATA-CUBE3 digitizers, recording continuous ground motion at a sampling rate of 100 Hz. The study was supplemented by two regional permanent networks, namely 16 stations from the Observatorio Vulcanológico *y* Sismológico de Costa Rica (OVSICORI) of the Universidad Nacional Autonoma de Costa Rica and 6 from the Red Sismológica Nacional (RSN) of the Universidad de Costa Rica (RSN, 2011) (Fig. 2b).

3.2. Event locations

Data processing began with a Short-Term Average/Long-Term Average (STA/LTA) routine to detect volcanic tremors (VTs) using P-to-S times ≤ 3 s, yielding 967 detections. From these, 24,681 phases (12,738 P- and 11,943 S-waves) were manually picked. Initial event locations relied on the 1D P-wave velocity model from Quintero and Kissling (2001). The *hypocentre* and *automag* algorithms (Lienert and Havskov, 1995), implemented in SEISAN (Havskov et al., 2010), handled phase picking and local magnitude (M_L) calculation, respectively; the latter used maximum amplitudes from Wood-Anderson filtered seismograms up to 600 km away. Seismic magnitude distribution appears in Fig. 3a.

The Wadati method (Wadati and Oki, 1933) identified useful V_P/V_S ratios of 1.74 to 1.8 for relocation, with 1.78 yielding optimal hypocentres locally. Magnitude of completeness (M_C = 1.8) was estimated via maximum curvature following Thompson and Reyes (2018) in the *GISMO* toolbox, alongside a b-value of 0.82 (small-to-large earthquake ratio). These parameters are shown in Fig. 3a, b. At this threshold, tectonic microseismicity (M < 0) evades STA/LTA detection.

The initial catalogue locations derived using regional 1D velocity models (Quintero and Kissling, 2001; Sallarès et al., 2000) were affected by artefacts resulting from local oversimplifications in the velocity structure. Hence, we derived a new local model by selecting reliable events (clear picks on ≥5 stations, RMS ≤0.9 s, errors <10 km), producing a 434 event subset (5459 P-phases). *Velest* (Kissling et al., 1994) was used to invert travel times for hypocentres and velocities, starting from the previously available velocity models (Quintero and Kissling, 2001; Sallarès et al., 2000), plus three arbitrary ones (constant 6 km/s; low/high P-velocity extremes with 2 km upper-crust and 5 km lower-crust steps; see Fig. 3b). After 450 runs (30 iterations each), the lowest-uncertainty structure (via *Hypoinverse* and *Velest*) was chosen, enhancing relocation accuracy for ITVC-recorded events (see Supplementary Fig. 3).

We present the original regional velocity models and high/low tests in Fig. 3b, the final 1D P wave velocity model in Fig. 3c and the distribution of earthquakes with depth in Fig. 3d. Values for S-wave velocities are calculated using the V_P/V_S of 1.78. Station corrections are also calculated at each station location (Fig. 4). This was achieved by inverting the initial data set considering new P- and S-wave travel times calculated using the new 1D velocity model. Fig. 3d shows that earthquake distribution with depth reduces significantly after 15 km, therefore we limit our local 1D velocity model to 25 km.

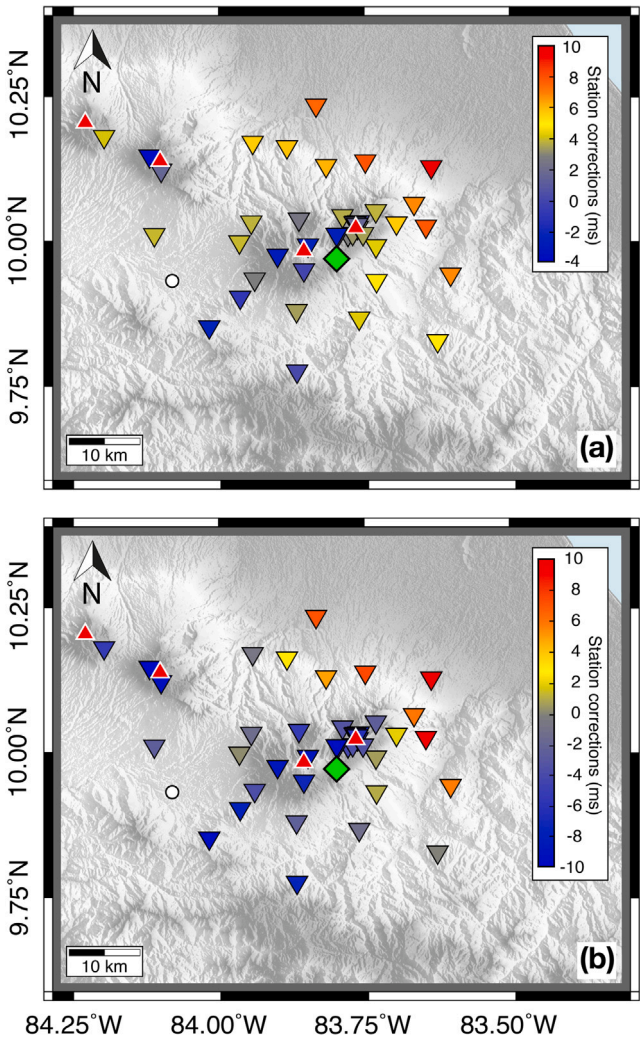


Fig. 4. Station Corrections. Station corrections from differences between **(a)** P and **(b)** S velocities calculated using the new 1D velocity model compared to that of the initial data. The difference is relative to the central ARA station (green diamond). Stations at lower altitude and away from volcanic edifices appear to have arrival times faster (red) then that of those located within the volcanic edifice and close to the crater summits (blue). Also, regions of higher topography also exhibit slower arrival times.

Hypocentres were refined via *HypoDD* double-difference relocation (Waldhauser and Ellsworth, 2000), employing the new velocity model (Fig. 3c) and recalculated P/S times. This scheme clusters diffuse seismicity along faults or contacts, minimising 1D model dependence in 3D space. Least Squares with QR-factorisation (LSQR) conjugate gradients used first arrivals and cross-correlations (Paige and Saunders, 1982; Waldhauser and Ellsworth, 2000), relocating 665 events. Most exclusions stemmed from <5 station coverage.

We calculated focal mechanisms using the *FPFit* program (Reasenberg, 1985) integrated in SEISAN. This uses first motion polarities of P-waves to find the double-couple fault plane solution and requires identifying at least six clear dilatational or compressional first motion arrivals. We use the first-arrivals to derive the azimuth and plunge for the principal stress directions of the catalogue. *FPFit* provides a critical analysis of the computed focal mechanisms. To achieve this, we calculate three different quality control measures, namely: (1) F-factor, an expression of the first motion polarity variances, (2) station distribution ratio (STDR), an assessment of the network arrangement on the received data, plotted on a focal sphere with respect to the

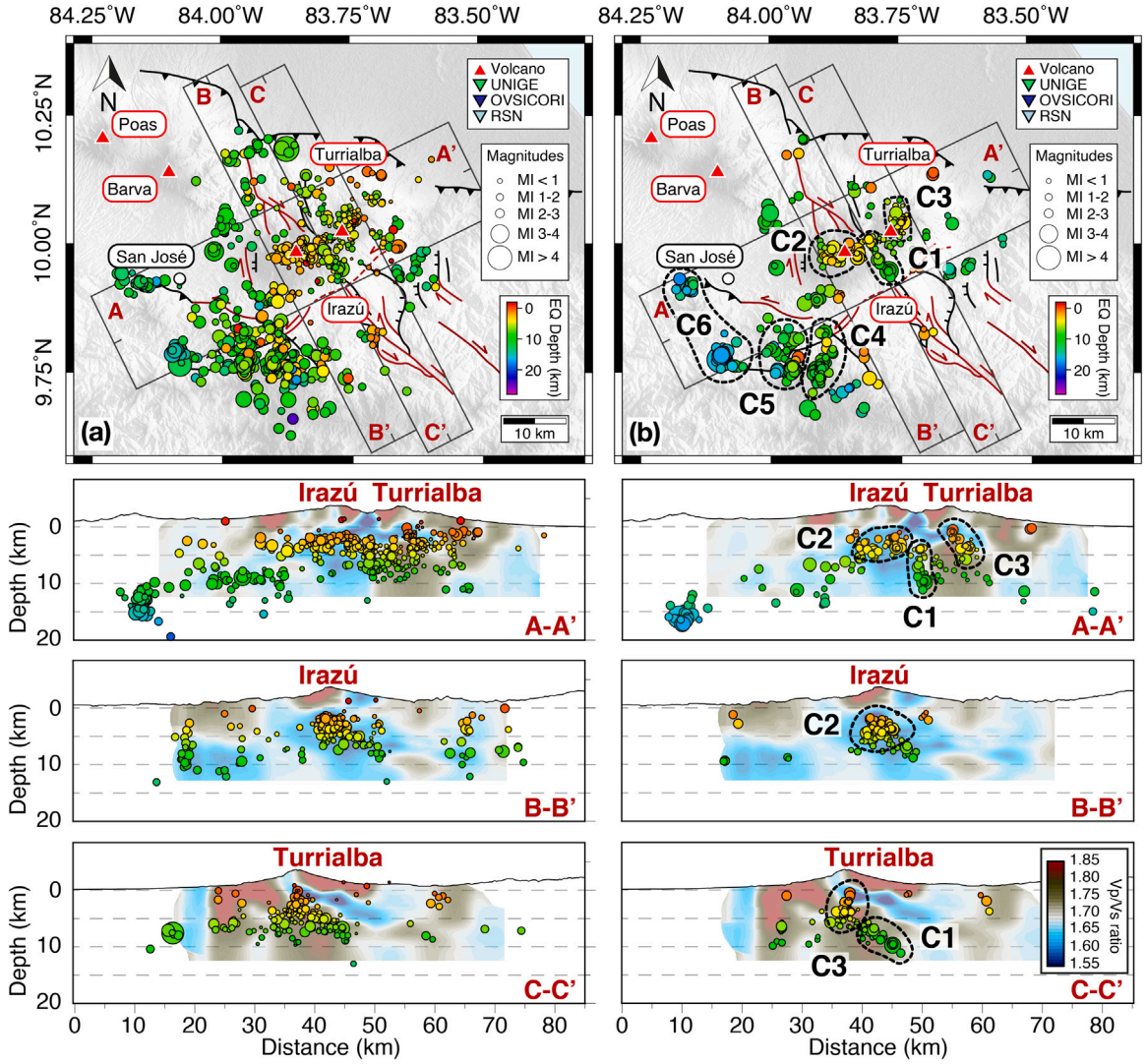


Fig. 5. Seismic Relocation. (a) Seismicity located with *HypoInverse* and the original, regional velocity models. **(b)** Seismic relocations after *HypoDD*. The main local tectonic features, labelled in more detail in Figs. 1 and 2 adapted from Denyer (2009), Montero et al. (2013b) and Alvarado et al. (2017) are also indicated on our maps. Cross sections along the axis of the ITVC (A-A') and perpendicularly for both Irazú (B-B') and Turrialba (C-C') are also plotted for both relocation methods. Local earthquake tomography results from Jiwani-Brown et al. (2024) are plotted below the relocation results. Volcanoes are represented by red triangles.

radiation pattern, and (3) errors in the strike, dip and rake, which we restricted to < 10°. Typical values of both the F-factor and STDR are usually less than 0.5, representing a good fit between picks and data proximity to the nodal planes (Reasenberg, 1985). Values greater than this were discarded.

4. Results

Velest-HypoDD comparisons are shown in Fig. 5 and reveal six clusters (numbered in b). Cluster 1 forms a linear shape equidistant from Irazú and Turrialba summits at 5–15 km depth. Shallow events (<5 km) concentrate in clusters 2–3 atop each volcano. Cluster 4 traces a sub-vertical line (5–10 km) perpendicular to Navarro fault. The neighbouring Cluster 5 spans 0–15 km on a sedimentary basin's southern flank, ≈10 km south of Irazú. Cluster 6 comprises two sub-clusters with deepest events (> 15 km) at ITVC's western edge.

We present the historical catalogue of focal mechanisms collected by the RSN in Fig. 6a and compare them to the local event solutions obtained during our temporary network deployment in Fig. 6b. The solutions of the historical catalogue and obtained solutions in this study indicate a large amount of both NW right-lateral and NE left-lateral strike-slip faulting for most activity, with M_L > 3. The largest event within the ITVC (M_L = 3.8) shows a reverse fault solution. Larger events (M > 4.0) recorded by our seismic network (Fig. 3a) are regional events and are not located in our study region. We present 3 clusters from the *HypoDD* relocations directly below the volcanic edifice. We locate this at 15 km north of Irazú volcano, close to the Guápiles thrust fault (Fig. 2a).

We analyse the focal mechanisms by evaluating the local stress field, which we report as the azimuth of the maximum horizontal stress, $S_H max$ (Supplementary Material, Table 2). This is calculated using the World Stress Map methodology (Heidbach et al., 2016), which accounts for the P, B and *T* axes associated with our focal mechanisms from which we can derive empirical values representative of the tectonic regime (divergent, convergent or transform) (Zoback, 1992). This presents an alternative expression of the principal stresses (σ_1, σ_2 and σ_3) (Fig. 7a). We also compute the azimuth and plunge of the principal stress axes for the three clusters (Fig. 7c - d) using the *StressInverse* package (Vavryčuk, 2014).

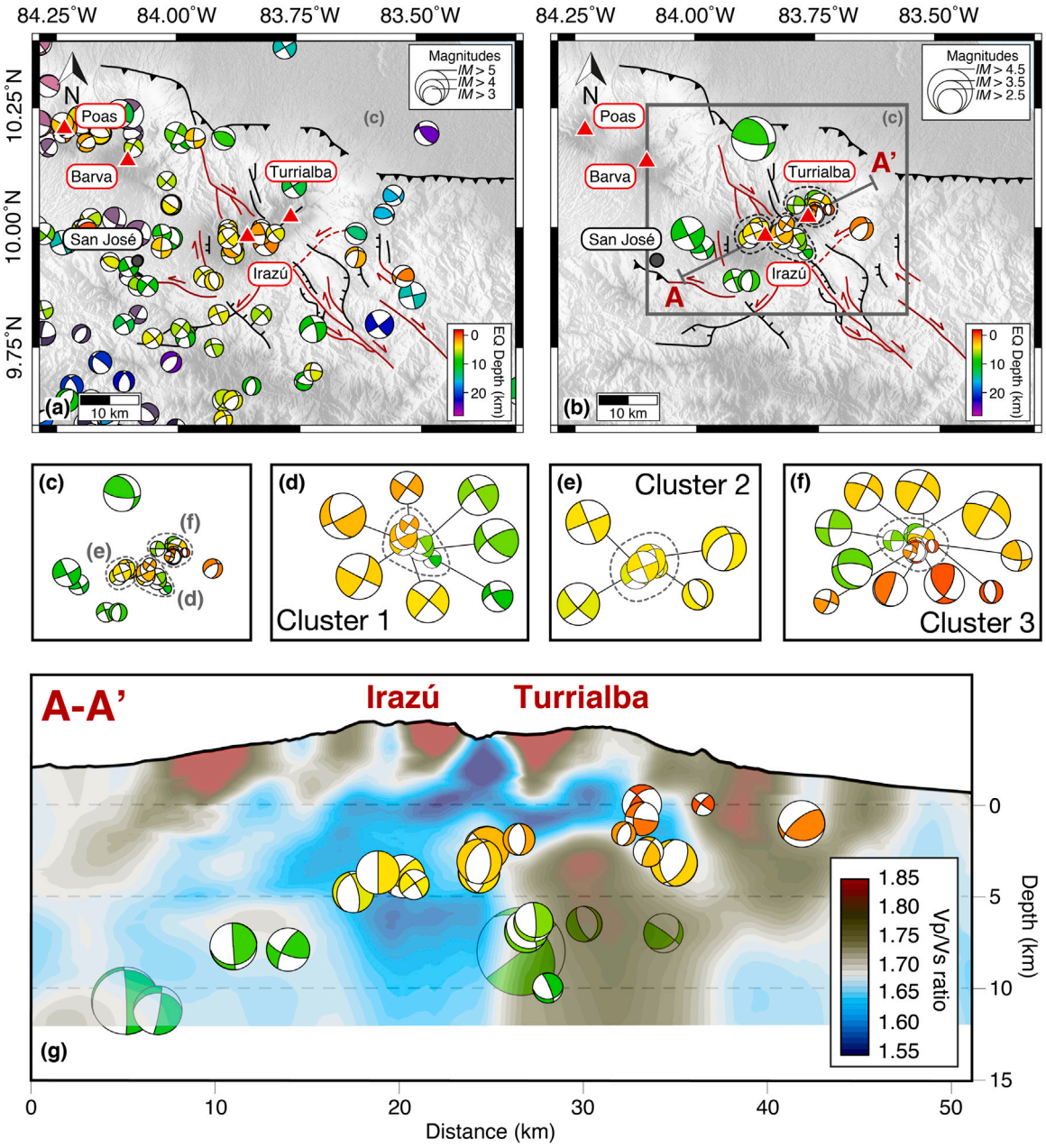


Fig. 6. Focal Mechanisms. Fault planes of the focal mechanisms, locations and magnitudes of seismic events recorded from **(a)** the local-scale historical RSN catalogue of $M_W \geq 4.0$ earthquakes and **(b)** the *HypoInverse* relocated, local-scale earthquakes recorded within the duration of this study. **(c)** Isolated catalogue with identified clusters of focal mechanisms. Zooms of the clusters are presented in **(d)**, **(e)** and **(f)**. **(g)** Focal mechanisms plotted in cross-section view with local earthquake tomography results from Jiwani-Brown et al. (2024) projected beneath the solutions. For this cross-section view, focal mechanisms are plotted at double the scale of Fig. 6b. The focal mechanisms for local-events were calculated using *FPFit* and manual polarity picking. Errors of dip, rake and fit were less than 10° for the 27 solutions plotted.

5. Discussion

Seismicity in volcanic arcs comprises a spectrum of source processes, but high-frequency volcano-tectonic (VT) earthquakes are generally attributed to brittle failure in the upper crust and can be treated using standard earthquake source analysis (Chouet et al., 2003; McNutt, 2005; Zobin et al., 2016; Neuberg, 2011; Wassermann, 2012; Chouet and Matoza, 2013; Saccorotti and Lokmer, 2021). In the ITVC, VT events therefore provide a direct window into how the brittle host rock responds to both regional tectonic forcing and magmatic and hydrothermal processes, complementing geodetic, tomographic and petrological constraints on the magmatic plumbing system (Jiwani-Brown et al., 2022, 2024; Alvarado et al., 2006; Di Piazza et al., 2019; DeVitre et al., 2019).

5.1. Multi-scale tectonic behaviour from seismicity within the ITVC

We present the main regional tectonic features recognised by previous studies in Fig. 8a. We segment each tectonic block, providing a visual representation of the relative effects that the Cocos-Caribbean subduction, CAFA northwest escape, and rotating Panama Microplate have on the diffuse deformation zone of the CCRDB. The proposed motions of these main tectonic blocks in Fig. 8a are based primarily

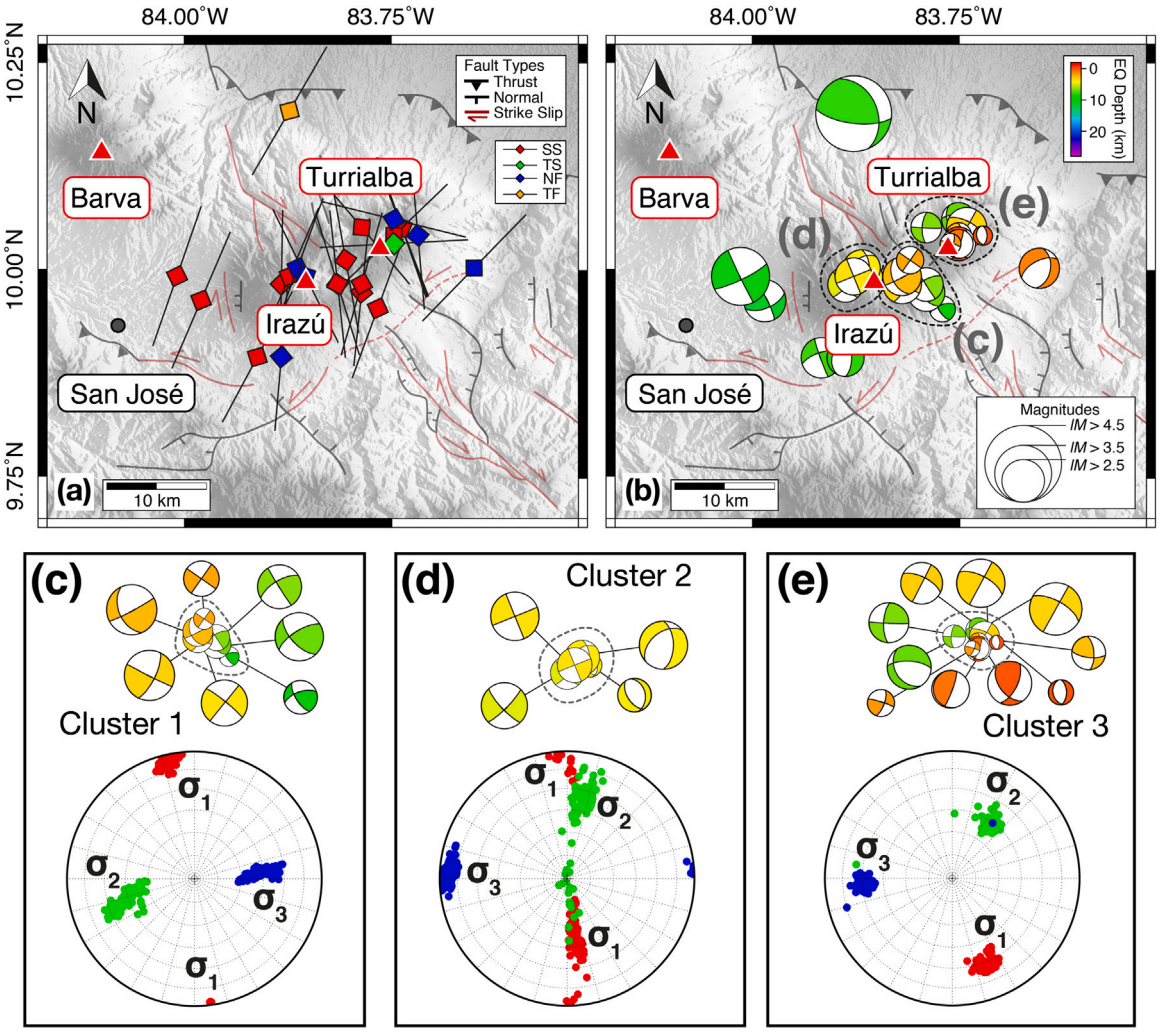


Fig. 7. Orientations of Principal Stress. (a) S_{Hmax} derived from focal mechanisms of volcano-tectonic earthquakes within the vicinity of Irazú and Turrialba volcanoes. Diamond symbols represent the stress regime, with lines showing the axial direction of these. Strike-slip (red), thrust with strike-slip (green), normal faulting (blue) and trust faulting (orange) regimes are indicated. **(b)** *HypoInverse* relocated, local-scale earthquakes from Fig. 6. **(c - e)** Identified clusters of focal mechanisms plotted with confidence limits of the principal stress directions calculated for each cluster with *StressInverse* (Vavryčuk, 2014).

on an independent GPS study (LaFemina et al., 2009), inverted Global Navigation Satellite System (GNSS) observations, explored in more detail by Kobayashi et al. (2014) and Carvajal-Soto et al. (2020) and modelling shear stresses at the base of the Caribbean Plate (Van Benthem and Govers, 2010). Plate motions relative to the Caribbean Plate indicate N-NW motions along the inferred western boundary of the CCRDB and a number of NW motions along the strike of the volcanic arc, evidence of the tectonic escape of the CAFA. Kobayashi et al. (2014) proposed a NE counter-clockwise motion of the PM at the eastern boundary of the CCRDB, within which deformation of these opposing motions is accommodated. Although the PB has an overall rotational motion, at the western limit deformation is predominantly defined geodetically and structurally parallel to the eastern boundary of the CCRDB, influenced by the Cocos Ridge collision (Van Benthem and Govers, 2010; Kobayashi et al., 2014). Van Benthem and Govers (2010) proposed that slab pull, frictional forces with surrounding plates and trench suction are the primary influences on the Caribbean Plate. This is coupled with low basal shear stresses, supporting the lack of asthenospheric flow acting on the regional plate dynamics.

López-Saborío and Carvajal-Soto (2024) used focal mechanisms compiled by Linkimer et al. (2021) to produce a regional scale interpretation of the stress field of Costa Rica, deducing that stress accumulation within Central Costa Rica attributed 60% of deformation due to coupling effects. They suggest that the regional stress regime in the upper 15 km of the CCRDB is dominated by strike-slip and transpressional kinematics, but our local high-resolution data reveal that normal faulting also plays a noticeable role at the volcano-scale, suggesting more complex fluid-driven and magmatic processes compared to the regional scale interpretation of López-Saborío and Carvajal-Soto (2024). This suggests that more complex dynamics such as mechanical loading, fluid-flow, temperature change and structural loading are at play. Fig. 7 shows the maximum horizontal stress orientations, derived from the focal mechanisms. SSW-NNE σ_{Hmax} (Fig. 7a) fits Cocos-Caribbean subduction and transtensional regime with a convergence azimuth of ≈ 25°. This is accommodated by a segmented, trench-parallel dextral strike-slip system that also generates pull-apart basins at step-over zones, analogous to the structures we infer beneath the ITVC.

Fig. 8b shows our proposed conceptual model. We combine our local-scale interpretations of structures with regional-scale tectonic interpretations and segments, inferred from the literature (Marshall et al., 2000; Carvajal-Soto et al., 2020; van der Laat et al., 2022; López-Saborío and Carvajal-Soto, 2024) that could accommodate the regional deformation. Given the presence of both of strike-slip and normal faulting at the local-scale (Fig. 6b), we suggest that this direction of maximum horizontal stress corresponds with the combination of right-lateral motion accommodated within the Atirro-Río Sucio fault system,

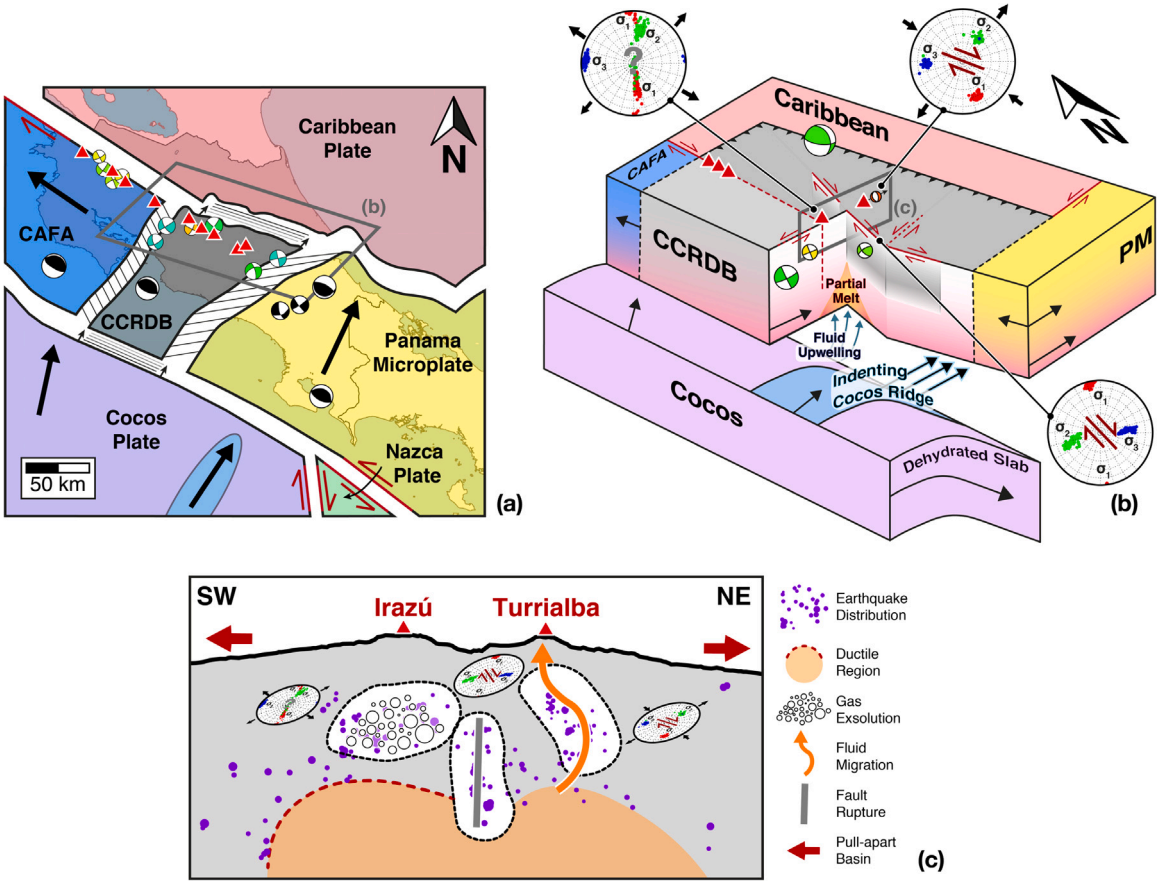


Fig. 8. Conceptual Tectonic Model. Proposed kinematic model to explain the multi-scale deformation due to oblique subduction and formation of tectonic blocks in a transtensional environment. **(a)** Regional-scale depiction of the main tectonic structures that act on the CCRDB. Black arrows indicate movements of the tectonic blocks with respect to the Caribbean Plate, modified after Carvajal-Soto et al. (2020). Red lines and arrows are the major strike-slip features. The eastern and western limits of the CCRDB are not defined boundaries and are instead inferred from the literature (Montero et al., 2013a; Carvajal-Soto et al., 2020; Jiwani-Brown et al., 2022). Lines from CCRDB connecting to the Central America Forearc Sliver (CAFA) and Panama Microplate are the proposed directions of deformation that occur within the CCRDB. Solid lines between the Cocos, Caribbean and CCRDB indicate the regions of subduction. Major earthquakes taken from Figs. 1 and 6a are also plotted. **(b)** Local-scale depiction of tectonic deformation within the ITVC. Red-solid lines show regions of strike-slip motion, and dashed lines are inferred continuations of this shear regime. Coloured gradients suggest the transition from shallow and intermediate crustal (≤ 15 km) to a deeper, subduction-dominant regime. Black arrows show simplified directions of motion for the tectonic blocks in this local-scale environment. Examples of the calculated focal mechanisms in this study from Fig. 6b are placed within the CCRDB. **(c)** Schematic subsurface structure of the ITVC based on the distribution of seismicity, focal mechanisms and stress inversions.

and left-lateral motion forming the pull-apart basin within which the ITVC has formed (Fig. 8b). It is important to note that this is limited to the number of resolved focal mechanisms. The NW-SE right-lateral regime intersecting the ITVC appears to be a resulting en-echelon structure. We propose that this formed due to a local tectonic processes, while also being influenced by the NW departing forearc sliver, acting on the western limits of the CCRDB. More NE-SW orientated maximum horizontal stresses are observed within the Turrialba volcanic zone (Fig. 7). The locations and kinematics of locally derived seismic events present evidence for intra-arc deformation acting on each volcano confirm that Irazú and Turrialba are subject to different degrees of regional tectonic shearing.

HypoDD relocated earthquakes reveals linear structures (Fig. 5b) implying unmapped faults linking to Río Sucio or Atirro-Turrique segments via the strike-slip Liebres continuity (Cluster 1, Fig. 6b Linkimer et al., 2018). Stress tensors show sub-vertical σ_1 (N350°), plunging $\sigma_{2,3}$ (30°), favouring NW right-lateral arc slip amid forearc escape and NNE compression (Alvarado et al., 2017; Montero et al., 2017; van der Laat et al., 2022; Montero and Alvarado, 1995; Montero et al., 2013b; Calvo et al., 2019). Cluster 2 is orientated perpendicular to Cluster 1 beneath Irazú (Figs. 5b, 6b,c), mixing strike-slip and normal mechanisms, which accommodate left-lateral transtensional fracturing. The alignment of these events suggests the existence of fracturing that supports the left-lateral motion. We propose that this cluster is associated with localised processes associated with Irazú volcano. However, Marshall et al. (2000) and Carvajal-Soto et al. (2020) suggest that similarly orientated motion could be accommodated within the internally deforming CCRDB, in the form of transtensional faulting. Cluster 3 lacks a clear linear structure but consists of NE-SW striking, transcompressional focal mechanisms at <5 km (σ_1 N170 to 180° Figs. 7d,e). However, the low number of resolvable focal mechanisms reduce the confidence of Cluster 2. Turrialba's en-echelon normal and strike-slip events suggest a pull-apart basin (third Río-Atirro segment Linkimer, 2005; Montero et al., 2013a; Soto, 1988; Linkimer, 2005).

We propose that the shallowest (< 5 km depth) seismicity is associated with left-lateral strike-slip deformation. This is a consequence of the complex combination of local-scale kinematics associated with the formation of a pull-apart basin striking parallel to the ITVC. This could also be influenced in part by the subducting Cocos Ridge, responsible for the cessation of volcanism, and overriding Panama Microplate tectonic block. Intermediate depth seismicity (5–15 km) is predominantly attributed to local-scale magmatic processes. However this could be kinematically linked to the departing forearc sliver and dextral shear along the volcanic arcs axis. These observations are not conclusive from

just our dataset, however the analysis of the historical seismic catalogue (López-Saborío and Carvajal-Soto, 2024) also supports a more locally-driven regime in the upper 15 km. Linkimer (2005) maps the complex faulting on both Irazú and Turrialba volcanic zones, reinforcing this transtensional environment instigated by this oblique subduction environment. This is supported by similar calculations made by Montero (1994) and Kolarsky et al. (1995).

5.2. Seismicity and subsurface structures

Having established the regional framework, we now examine how the local seismicity and subsurface structures reflect these distinct tectonic regimes. Volcano-magmatic systems form in the intermediate-to-upper crust (<15 km) through brittle host rock interactions with intruding juvenile magmas (Cashman et al., 2017; Sparks et al., 2019; Edmonds et al., 2019; Acocella, 2021). Petrological and geochemical studies (e.g. magma evolution and mixing (Alvarado et al., 2006)), isotope analysis and strong modelled seismic velocity anisotropy values (Hoernle et al., 2008; Gazel et al., 2009), and trace-element data (DeVitre et al., 2019; Gazel et al., 2019)) reveals a 10 km ITVC reservoir (Alvarado et al., 2006; Di Piazza et al., 2015; de Moor et al., 2016; Di Piazza et al., 2019; DeVitre et al., 2019), while multi-disciplinary spatial temporal deformation (Muller et al., 2024) and ambient noise tomography identifies a shared 6 km m.b.s.l. mush zone (Jiwani-Brown et al., 2022).

We infer the subsurface structure of the ITVC in Fig. 8c. *Velest* relocations (Fig. 5a) show a diffuse pattern of seismicity in the depth range 0–10 km, with the number of events decreasing below 10 km depth and deepening at 50 km along A-A'. *HypoDD* clustering (Fig. 5b) clarifies a sharper aseismic gap (Irazú > 5 km, Turrialba > 8 km), matching high/low V_P/V_S magmatic anomalies and mush zones (Jiwani-Brown et al., 2024, 2022). Cluster 1 (Fig. 5b) crosses this zone, resembling the Liebres fault (5–15 km), which was activated by the M_L5.5 Capellades quake (Linkimer et al., 2018; Soto et al., 2009), and is thought to have been triggered by fluid and stress permutations (van der Laat et al., 2022; Muller, 2018; Battaglia et al., 2019).

Melt-rock contacts and magma migration within the crust (e.g., upward melts (Paterson et al., 2011), sills (Sparks et al., 2019)) can trigger volcano-tectonic earthquakes (VTs) and change the stress field, inducing brittle failure of the country rock. These zones can occur at, or surrounding the brittle–ductile transition common in volcanic environments. The proposed ductile region beneath Irazú and Turrialba is outlined by a number of seismic clusters, and has previously been interpreted as melt pocket roofs with fluid exsolution and fracturing (Sparks et al., 2019; Acocella, 2021; Neuberg et al., 2006). Cluster 2 spans 5 km laterally (radial to 10 km) and is consistent with post-2019 deflation of approx 5 mm/yr (OVSICORI-UNA, 2020; van der Laat et al., 2022). This cluster could be explained by volume changes from recharge and depletion (Battaglia et al., 2005; Syracuse et al., 2016; Christopher et al., 2015) that drive boundary fracturing. Cluster 3 forms sub-vertical paths (A-A' 60 km, C–C' 35 km) suggesting fluid ascent amid Turrialba unrest, bridging surface and 6 km low V_P/V_S anomalies and summit degassing (Epiard et al., 2017; Montero-Pohly, 2021b; Gazel et al., 2021; Conde et al., 2014; de Moor et al., 2016; Jiwani-Brown et al., 2024; Muller et al., 2024).

Clusters 4 and 5 align along the EW-trending Navarro fault. Marshall et al. (2000) and Montero (2001) define this as a left-lateral structure within the broad region of conjugate faulting associated with the CCRDB. The western limits of this fault branch off into a more complex system of undefined faults, and are intersected perpendicularly by the normal slipping Orosi fault. We propose that Cluster 5 appears to fit with previously defined faulting at the surface. Our locations for Cluster 4 produce a linear structure, intersecting at the same point as the Navarro and Orosi fault. However, locations are orientated north–south, equidistant from the Navarro and Orosi faults. We propose that this could represent the formation of a new, previously undiscovered fault. However the low magnitudes of the events, and lack of focal mechanisms for this cluster mean we cannot deduce more than just the spatial extent and not the kinematics.

We observe the deepest seismicity (> 15 km) in Cluster 6. These exist at the limits of Aguacaliente and Navarro fault systems (Montero et al., 2013b,a; Montero-Pohly, 2021b). The Aguacaliente system has been responsible for previous destructive earthquakes (Cartago 1910) and has been characterised, similarly to the Navarro fault, as a predominantly left-lateral E-W striking fault. We notice that the *Velest* relocations (Fig. 5a) are distributed more linearly when compared to the *HypoDD* for this cluster. As these clusters are located outside the limits of our network, the resulting *HypoDD* cluster could be an artefact of the methodology where results are improved within the limits of the network.

5.3. Local and global perspective, and limitations

Volcanic arc rheologies create weak zones in strong crust (Cailleau et al., 2007; Cashman et al., 2017), yielding arc-parallel shear dominance as in Nicaragua (Cailleau et al., 2007). In the Nicaraguan segment, this margin-parallel shear is accommodated by bookshelf faulting on inherited, margin-normal sinistral faults (Higgins et al., 2022, 2024). These are spatially confined to the mechanically stronger crust between volcanic centres contrasting with the arc-oblique conjugate fault network that directly intersects the ITVC within the CCRDB. The interplay of the Cocos and Caribbean Plates in Costa Rica has resulted in intra-arc deformation zones (Montero and Alvarado, 1995; Montero et al., 2013b; Alvarado et al., 2017; Carvajal-Soto et al., 2020); Cocos-Caribbean oblique subduction and resulting strain partitioning drives northwestward-directed forearc sliver motion imparted dextral shear across the volcanic arc, and Cocos Ridge collision drives northeast-directed shortening and east-northeastward escape of the Panama (Montero et al., 2013b; Alvarado et al., 2017; Montero et al., 2017; Carvajal-Soto et al., 2020). Petrone et al. (2006) observed a lithological bi-modality in the San Pedro-Cerro Grande volcanic complex. The structural compartmentalising of a volcanic system in this way can facilitate the ascent of geochemically diverse magmas through independent crustal pathways. Although this is not yet documented at the ITVC, distinct magmatic series emplaced along fault-controlled plumbing systems, like that observed by Petrone et al. (2006), could occur without interaction or magma-mixing, something that could arise from independently plumbed systems like that observed from previous geophysical imaging studies (Jiwani-Brown et al., 2022, 2024).

Seismic and geodetic data (Marshall et al., 2000; Bird, 2003; Lewis et al., 2008; LaFemina et al., 2009; Correa-Mora et al., 2009; DeMets et al., 2010; Argus et al., 2011; Montero et al., 2013b; Kobayashi et al., 2014; Alvarado et al., 2017; Montero et al., 2017; Carvajal-Soto et al., 2020) shows that in NE Costa Rica, right-lateral shear motion from oblique subduction between the Cocos and Caribbean Plate is accommodated by the forearc sliver. This presents a tectonic environment subject to right-lateral strike-slip motion from the NW-SE departing CAFA sliver, and left-lateral shear from the northeast escape of the Panama Microplate, also observed in several other studies (Montero et al., 2013b; Alvarado et al., 2017; Montero et al., 2017; Carvajal-Soto et al., 2020). This complex regional motion needs to be accommodated, and our data appear to suggest this. Historical shallow (≤ 20 km, M>4) focal mechanisms confirm strike-slip abundance (Fig. 6a (RSN, 2011)).

At a more regional scale, the pattern of arc-oblique faulting accommodating intra-arc deformation shares parallels with the Andean Southern Volcanic Zone, where reactivated WNW-striking inherited basement faults control the distribution of volcanic and hydrothermal systems and are similarly misoriented relative to the regional stress field yet activated by overpressurised magmatic fluids (Singer et al., 2008; Cembrano and Lara, 2009; Pearce et al., 2020). The Great Sumatra Fault is also an example where oblique convergence,

of the Australian–Eurasian Plate, drives ≈15 mm/yr of arc-parallel forearc sliver motion (Sieh and Natawidjaja, 2000; Bradley et al., 2017; Acocella et al., 2018).

Our relocations, based on a new minimum 1D velocity model and double-difference (*HypoDD*) methods, sharpen the seismicity patterns compared to locations obtained with regional 1D models. The resulting clusters reveal discrete structures beneath and around the ITVC that were not apparent in the original catalogue, and provide a basis for linking shallow intra-arc deformation to the CCRDB fault network. However, limitations can arise from using a 1D model for 3D distributions of seismic events. Fig. 4 indicates that the distribution of station corrections are consistent with a combination of topography and the orientation of NW-SE strike-slip faults. Also the 3D tomographic images from ANT (Jiwani-Brown et al., 2022) and LET (Jiwani-Brown et al., 2024) show a similar trend in the seismic velocities, compatible with that of the station correction observations.

We assume double-couple focal mechanisms for these VT events, which is appropriate given their small magnitudes and the limitations of our dataset. Given the local setting, we cannot exclude that some of our inversions could have required a non-double-couple approach. In principle, full moment tensor inversions could resolve non-double-couple components and volumetric contributions related to fluid or magma movement. However, most of our events are micro-earthquakes ($M_W < 2$) with low signal-to-noise ratios at the low frequencies required for robust full-waveform inversions (Fojtíková et al., 2010; Fojtíková and Zahradník, 2014; Carvalho et al., 2016). Unlike studies targeting aftershock sequences of well-recorded moderate events, we lack suitable reference events and dense near-field coverage to stabilise such inversions. We therefore focussed on first-motion polarity solutions, which are sufficiently reliable to extract the dominant faulting styles and stress orientations within the brittle crust. Given the distance of our locations from the main volcanic centres we assume that the solution using polarities is a robust approach for describing the overall seismicity we present.

6. Conclusion

We investigated the kinematics of the region around the Irazú–Turrialba volcanic complex, using a temporary network of broadband seismic stations that integrated the permanent Costa Rican network. We examine the main local-scale tectonic features and investigate the influence of regional-scale tectonic motions. We relocated 967 seismic events and constrained 27 focal mechanisms, from April 2018 and March 2019.

The earthquakes recorded during our experiment highlight boundaries of subsurface volcanic structures, and a complex tectonic setting, interacting with an active volcanic system. Distributions and clustering of seismicity localised beneath the Irazú volcano suggest the upper limit of a ductile region where we record no seismicity, and normal faulting provides evidence of decompression, correlated with observed subsidence, just above the inferred magmatic reservoirs. A more tightly distributed cluster of events below Turrialba suggests the upward migration of fluids above the active magmatic region, fuelling volcanic activity. Seismic events also show further evidence of a well-developed fault, intersecting the ITVC. Orientations of these events suggest a NW-SE striking right-lateral fault, connecting the Río-Sucio and Atirro–Tucurrique fault segments, which are obliquely orientated relative to the slip accumulated along the Central America Volcanic Arc. Focal mechanisms for the cluster below Turrialba indicate a NE-SW orientated system, with a combination of strike-slip and normal faulting, characteristic of a pull-apart basin formation.

We suggest that deformation associated with Irazú volcano is predominantly locally-driven, however is likely to be influenced by the regional-scale NW departure of the Central America Forearc Sliver. Turrialba has multi-scale influences from both independent local deformation, and the departure of the Panama Microplate, due to the shallowing subducting Cocos Plate and morphologically complex and buoyant subducting Cocos Ridge. Our local scale seismicity study revealed the active tectonic features within the ITVC during a one-year period using volcano-tectonic events. We were also able to constrain boundaries between brittle–ductile transitions at depth, while further investigations of other volcanic seismicity could reveal deformation within the volcanic system due to the migration of magmatic fluids.

CRediT authorship contribution statement

Elliot Amir Jiwani-Brown: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Francisco Muñoz-Burbano:** Writing – review & editing, Visualization, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Juan Porras:** Writing – review & editing, Methodology. **Javier Francisco Pacheco:** Writing – review & editing, Supervision, Project administration, Formal analysis, Data curation, Conceptualization. **Mauricio M. Mora:** Writing – review & editing, Project administration, Investigation, Data curation. **Matteo Lupi:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Elliot Amir Jiwani-Brown reports financial support was provided by University of Geneva. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.tecto.2026.231408.

Data availability

Data from all stations (temporary and permanent) for the acquisition period of April 2018 to March 2019 can be found under the DOI: https://doi.org/10.5281/zenodo.4912497.

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